

Bhavya Sahithi Indukuri

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer with nearly 3 years of experience in application development, including 1 year focused on Generative AI and Retrieval-Augmented Generation (RAG). Worked on RAG-based document question-answering systems using LangChain, OpenAI, Pinecone, and FAISS, covering document processing, embeddings, vector search, hybrid retrieval, reranking, and evaluation. Previous experience includes Python backend development, REST APIs, SQL, data processing, unit testing, and debugging.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Generative AI & ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search, Hybrid Search, Reranking, RAG Evaluation

Frameworks & Libraries: LangChain, LangSmith, Hugging Face Transformers, RAGAS, BM25

Vector Databases: Pinecone, FAISS, ChromaDB

Tools & Platforms: Streamlit, Git, Visual Studio Code

EXPERIENCE

Cognizant Technology Solutions - Bengaluru, Karnataka

May 2024 – Apr 2025

AI Engineer

- Transformed an FAQ-based chatbot that could answer only predefined questions by adding RAG, enabling it to retrieve answers from internal company documents and helping employees resolve more issues without contacting support.
- Built the retrieval pipeline using Python, LangChain, OpenAI, Pinecone, and FAISS, indexing 1,300+ internal documents resulting in 16,700+ chunks for semantic search.
- Developed document ingestion pipelines for PDF, DOCX, HTML, CSV, and scanned documents, including OCR, text cleanup, metadata extraction, deduplication, and chunking.
- Added BM25 and vector search with metadata filtering and reranking, improving Recall@5 from roughly 0.71 to 0.87 on the evaluation dataset.
- Evaluated retrieval and generated answers using Recall@5, MRR, faithfulness, and answer relevance, and used the results to tune chunking and retrieval settings.

Python Developer

Nov 2022 - Apr 2024

- Developed Python REST APIs for JPMC payment transactions, validating transaction IDs, account details, amounts, and processing dates before storing records in SQL.
- Built a Python and SQL reconciliation job to compare JPMC payment records with settlement data, using transaction IDs and amounts to identify missing, duplicate, and mismatched transactions.
- Added failed-transaction handling to capture validation and API errors in SQL, retry eligible transactions, and test common cases such as duplicate records, missing fields, and amount mismatches.

Intern

Mar 2022 – Aug 2022

- Worked on Python and Django backend features, including REST API changes and SQL queries for retrieving and updating application data.
- Fixed bugs, tested backend changes, and worked with Git for version control while learning the team's development and code review process.

PROJECTS

RAG Product Q&A Chatbot

- Developed a RAG-based product Q&A chatbot that retrieves relevant information from product data and uses an LLM to generate context-grounded responses.
- Implemented the RAG pipeline including document processing, text chunking, embeddings, vector retrieval, and LLM-based response generation.
- Designed the system to provide accurate, context-aware answers based on retrieved product information.

Agentic AI Workflow (LangGraph)

- Built an Agentic RAG system with LangGraph and LangChain, enabling LLM agents to dynamically select and use retrieval tools based on user queries.
- Integrated FAISS vector search, Hugging Face embeddings, retrieval tools, document grading, query rewriting, and contextual response generation.
- Designed a multi-agent orchestration workflow where planner and executor agents collaborate to decompose and solve complex multi-step tasks.
- Enabled the agent to grade retrieved documents and rewrite queries when relevant context is insufficient, supporting iterative retrieval and more relevant response generation.

EDUCATION

Master of Science - Computer Science and Engineering

Jan 2026 - present

University of North Texas, Denton, TX

Bachelor of Technology - Computer Science and Engineering

Aug 2018 – May 2022

Sri Venkateswara College of Engineering, Tirupati, Andhra Pradesh, India